

Received 17 March 2025, accepted 20 May 2025, date of publication 28 May 2025, date of current version 9 June 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3574547

RESEARCH ARTICLE

Preventing Spurious Interactions: A New Inductive Bias for Accurate Treatment Effect Estimation

ROGER PROS^{ID} AND JORDI VITRIÀ^{ID}

Departament de Matemàtica i Informàtica, Universitat de Barcelona, 08007 Barcelona, Spain

Corresponding author: Roger Pros (roger.pros@ub.edu)

This work was supported in part by the Ministerio de Ciencia, Innovación y Universidades under Project PID2022-136436NB-I00; and in part by the Generalitat de Catalunya under Project 2021SGR 01104. The work of Roger Pros was supported by an Industrial Doctorate Grant funded by Generalitat de Catalunya of Agència de Gestió d'Ajuts Universitaris i de Recerca (AGAUR) under Grant 2021 DI 41.

ABSTRACT In recent years, the use of non-linear machine learning techniques to estimate treatment effects from observational data has gained significant attention. Many of the most effective methods incorporate specific algorithmic inductive biases that encode causal information, thereby enhancing the precision of treatment effect estimates. In this paper, we explore and categorize several of these models based on the biases they address, and we introduce the prevention of spurious variable interactions as a new inductive bias that serves as a central focus of our study. This novel inductive bias, not previously considered in this area, aims to further improve the accuracy of treatment effect estimation. Additionally, we propose a practical method that utilizes knowledge of the data generation processes to restrict variable interactions to those defined by the mechanisms of the underlying causal model. This strategy integrates seamlessly with many of the leading neural network-based models. Our results show that these constraints significantly enhance performance, setting new state-of-the-art benchmarks in treatment effect estimation. Furthermore, we show that our method remains robust even when the causal model is imperfect and that incorporating partial causal information yields better results than disregarding it entirely.

INDEX TERMS Algorithmic inductive biases, neural networks, treatment effects estimation, variable interactions.

I. INTRODUCTION

Quantifying the effects of interventions or treatments on outcomes and predicting potential outcomes under various hypothetical scenarios are fundamental objectives of causal machine learning [19]. These goals are integral to decision-making across a broad spectrum of domains, including medical trials, economic policy analysis, evaluations of educational programs, social science research, and, more recently, artificial intelligence [35].

A major challenge in addressing these tasks lies in the inherent uncertainty surrounding potential outcomes under different interventions, a concept widely known as the “Fundamental Problem of Causal Inference” [17]. This challenge arises from the fact that, for any given instance, only the outcome corresponding to the treatment or intervention that was actually applied is observable, while

the counterfactual outcome—what would have occurred under a different treatment—remains unobservable. This limitation poses a significant challenge to accurately measuring causal relationships.

As a result, conventional data-driven approaches to estimating causal quantities, such as the treatment effect and its associated metrics (e.g., Average Treatment Effect (ATE) and Conditional Average Treatment Effect (CATE)), are often prone to substantial biases. These biases stem from the lack of direct counterfactual information, making it difficult to disentangle causal effects from confounding influences or noise in observational data. Overcoming this challenge is a central focus of causal inference research, driving the development of methods and models designed to better disentangle confounding influences and improve the reliability of causal effect estimations.

As a result, recent years have seen growing interest in developing techniques that adapt flexible machine learning models for estimating causal quantities [10]. This field,

The associate editor coordinating the review of this manuscript and approving it for publication was Alba Amato^{ID}.

in which neural networks play a pivotal role [36], [37], [39], [41], serves as the central focus of our paper. In this work, we provide a comprehensive review and classification of these models based on their algorithmic inductive biases — the assumptions that guide a model in making predictions or generalizations from its training data. Additionally, we investigate the interplay between these inductive biases and the different forms of biases that must be addressed to ensure accurate estimation of treatment effects.

A key focus of this paper is the identification and mitigation of a commonly overlooked bias in the literature: the presence of spurious interactions between covariates in models. Often, there is prior knowledge about the underlying data generation process, and this information can be utilized to characterize these interactions, enhancing the accuracy and robustness of causal inference. Prior knowledge can range from having a comprehensive or partial understanding of the causal graph to being aware of specific interactions between variables. In all these cases, this knowledge constrains the set of variable interactions that are consistent with the causal structure of the data. Leveraging this knowledge can significantly enhance the development of more effective models, as considering interactions not supported by the underlying causal model can create correlational shortcuts that distort the measure of the causal quantity estimated by models, especially when data is scarce. To address this source of bias, we propose a method, **Spurious Interactions Prevention (SIP)**, which can be applied to almost any existing neural network model designed to estimate treatment effects. Our experiments show that integrating these constraints into neural network estimators significantly enhances performance, setting new state-of-the-art benchmarks in treatment effect estimation. Furthermore, we demonstrate that our method remains robust even when the causal graphs are imperfect and that leveraging partial causal information is more effective than disregarding it entirely.

The rest of this paper is organized as follows. In Section II, we define the problem and its main components. In Section III, we describe the proposed approach. In Section IV, we present the research questions and the experiments we conducted to evaluate our method and compare it with other state-of-the-art estimators. In Section V, we discuss the results and analysis of our experiments and suggest some possible directions for future research. In the methods section, we detail the implementation of the proposed approach, outline the experiments conducted, and provide a review of related work in causal effect estimation.

II. PROBLEM DEFINITION

Our primary goal is to estimate an important causal quantity from observational data: the **causal effect (CE)** of a treatment variable T on an outcome variable Y . Estimating the causal effect requires disentangling the true causal relationships from confounding factors that may obscure or bias the observed associations in the data. This task is particularly challenging in observational settings, where the treatment assignment

is not randomized, and variables influencing both T and Y can introduce spurious correlations. Robust estimation of the causal effect, therefore, requires advanced techniques capable of addressing such complexities, including accounting for hidden confounders and utilizing domain knowledge when available.

We assume that the underlying causal relationships between these variables can be represented by a **Causal Graph** [20] that involves these variables and possibly some other variables. A causal graph is a directed acyclic graph (DAG) where each node is a variable and each edge represents a direct causal effect. We assume that the causal graph satisfies the causal edges assumptions, which means that all parents are causes of their children. We denote the causal graph by $\mathcal{G}$, the set of parents of node x_i by $Pa(x_i)$, its children by $Ch(x_i)$ and its ancestors by $An(x_i)$.

For a causal graph representing a data generation process, the joint distribution of the variables involved can be expressed in the following factorized form:

$$p(X) = p(x_1, \dots, x_k) = \prod_{j=1}^k p(x_j \mid Pa(x_j)) \quad (1)$$

In this setting, and according to the **Independent Causal Mechanisms (ICM)** principle [35], the generative model of the variables of the causal model consists of a set of k independent modules. These modules function autonomously, without sharing information or exerting influence on each other. Consequently, the conditional probability distribution of each variable x_j given its causal factors, $p(x_j \mid Pa(x_j))$, remains unaffected by and independent of other mechanisms in the model.

The ICM principle can be leveraged in the design of machine learning models by focusing on their representation in terms of causal conditionals, rather than relying on conditionals derived from alternative factorizations [29]. Causal conditionals align with the natural independence structure of the underlying data-generating process, preserving the independence of mechanisms. This ensures that changes in one mechanism, such as a causal factor, do not inadvertently affect others, making models more robust and interpretable. Additionally, models based on causal conditionals are better equipped to generalize across different settings, such as interventions or shifts in data distribution. This is because the learned mechanisms reflect stable causal relationships that remain consistent even when the context changes. By isolating the effects of specific variables, these models also reduce confounding biases that may arise from spurious correlations, leading to more accurate estimations of causal effects.

Additionally, the modular nature of causal conditionals allows each component to be learned and updated independently, making models more flexible and adaptable to changes, such as incorporating new variables or refining specific components without requiring retraining of the entire system. Furthermore, representing models in terms of causal conditionals enhances interpretability by providing insights

TABLE 1. Types of biases in causal inference with observational data.

Bias Type	Description
Identification biases:	
Exchangeability bias	Bias derived from the existence of hidden confounders that impact the identification process.
Positivity bias	This assumption can be violated if data for a specific subpopulation is not collected.
Consistency bias	Bias derived from the violation of the consistency assumption: if the treatment is T , then the observed outcome Y is the potential outcome under treatment T .
Non-interference bias	Bias derived from the effect of one individual treatment on another individual. The existence of this bias is particularly common in network data [23].
Estimation biases:	
Propensity bias	A causal model must generalize from behavior under one set of conditions to behavior under another set [32]. This bias appears when the performance of the model differs between distinct treatments due to a difference in the distribution of the data.
Relevance of treatment	Bias that appears especially in high-dimensional settings. Models tend to neglect the role of the treatment and its effect is expressed through correlated features.
Functional form	Task performance can be affected by the functional form in which it is modeled. Linear tasks are better modeled by linear models, and monotonic tasks by monotonic models.
Variable interactions	Spurious variable interactions do not explain any causal relationship between the variables, but a model might use them as a correlational shortcut, modifying the effect of causal relationships.
Data efficiency	A model might not make an efficient use of the information present in the data.
Missing data	Bias derived from the lack of availability of some data.
Measurement error	Bias derived from measurement error on some variables.

into the data-generating process, making predictions and model behavior more explainable.

In many real-world scenarios, the causal graph or the complete set of Independent Causal Mechanisms (ICMs) may not be entirely known. However, there is often expert knowledge available about specific interactions between variables within certain causal mechanisms. For instance, in an environmental study investigating the effects of pollutants on air quality, researchers may lack comprehensive information about all the chemical reactions occurring in the atmosphere. Nevertheless, they may be aware that certain pollutants interact with specific atmospheric compounds, while others do not.

Having defined a causal quantity (CE) and a causal graph ($\mathcal{G}$), the next step is to assess if this quantity can be uniquely inferred from data. This is called the identifiability process. In the case of CE, to ensure identifiability we have to fulfill several **causal assumptions**: having independent and identically distributed data, consistency, no interference, positivity, and unconfoundedness [17].

Unconfoundedness is a critical concept in causal inference. It refers to the condition where all variables that influence both the treatment and the outcome have been measured and accounted for in the analysis. Under this assumption, the treatment assignment is effectively random given the observed covariates. The **backdoor criterion** [31] provides a method to identify a set of variables X that, if conditioned upon, will satisfy unconfoundedness. This set of variables is called the **adjustment set**. In other words, the backdoor criterion helps identify the covariates needed to control for confounding.

In this work, we will focus on causal effects that can be identified using the backdoor criterion, as this is the most common scenario in applied settings where we have only access to observational data.

Finally, if we have access to a valid adjustment set X , it can be shown that the causal effect of a binary treatment T on Y can be estimated by this statistical estimand:

$$\mathbb{E}_X[\mathbb{E}[Y|T = 1, X] - \mathbb{E}[Y|T = 0, X]]. \quad (2)$$

This estimand closely resembles the commonly used target in generic machine learning prediction tasks, $\mathbb{E}[Y|T, X]$. However, the model selection processes and inductive biases effective for one may not be applicable to the other [9]. As a result, having an identified expression alone might be insufficient for computing accurate and robust causal effects. The identified estimand can still face challenges during estimation, potentially leading to complications that make non-asymptotic estimation infeasible [25].

In the literature, biases in causal inference are broadly categorized into two main types: **identification biases** and **estimation biases** (see Table 1).

Identification biases occur when the assumptions required to correctly identify causal relationships are violated. These assumptions are fundamental for linking observed data to the underlying causal structures. For example, violations of causal assumptions such as unmeasured confounders, incorrect causal graph specifications, or inappropriate model assumptions can lead to errors in identifying the true causal effect.

On the other hand, estimation biases arise during the process of estimating causal effects from the data, even when the causal assumptions necessary for identification are satisfied. These biases are often caused by practical limitations, such as finite sample sizes, model misspecifications, or noise in the data. Estimation biases can also result from the choice of the estimation method or its inability to fully capture the complexity of the underlying causal mechanism.

Both types of biases significantly impact the accuracy and reliability of causal effect estimation. Identification biases may prevent the recovery of the true causal effect entirely, while estimation biases can lead to imprecise or inconsistent results, even when the causal structure is well understood. Addressing both is critical to ensure robust and trustworthy causal inference, requiring careful consideration of the assumptions underlying causal models and the methods used to estimate causal effects.

A. CAUSAL EFFECTS ESTIMATION: STATE OF THE ART

In this section, we review some of the most effective machine learning methods for estimating Eq. 2 and discuss how they address identification and estimation biases.

Two of the most straightforward approaches for estimating Eq. 2 are the S-Learner and the T-Learner [21]. S-learner is a method that trains a single model with the treatment variable as one of the features, along with other covariates. This model is then used to predict outcomes under both scenarios - with treatment and without treatment. The main disadvantage of S-learner is that it can bias the estimate of the treatment effect towards zero, especially when the feature space is large. Unlike the S-learner, the T-learner splits the data by treatment and fits a model for each group. This addresses the issue of failing to recognize a weak treatment variable, yet it may still be prone to data efficiency bias.

S-learner and T-learner are among the simplest causal effects estimators; however, apart from these approaches, more sophisticated methods have also been developed. Most of the best performing methods rely on the concept of **representation learning** [1]. They use neural networks with an architecture that can be understood in two stages: pre-representation and post-representation. The first creates a representation from the input data and the second takes over from the representation to the desired output. Explicitly dealing with representations is interesting because they can express many general priors about the world around us. These priors can be used to incorporate inductive biases to improve the causal inference task. For example, TARNet [36] branches off into two separate heads for each treatment group to prevent the model from disregarding T . On top of that, Dragonnet [37] adds a third head to the TARNet architecture to predict propensity $g(X) = P(T = 1|X)$. This approach reduces the bias caused by having different modeling performances for each value of the treatment. BCAUSS [39] uses the same architecture as the Dragonnet, but modifies the loss function so that it is more robust to positivity violations. CFRNet [36]

and BNN [18] create a balanced representation of the data for both treated and untreated observations. Similarly, importance sampling weights can be added to the CFRNet to alleviate the problem of selection bias [14]. HINITE [23] addresses no interference assumption violations, and in [6] the authors create a representation that encodes prior information about the structure of causal effects.

Other works have suggested inductive biases that are challenging to link with a particular source of error. For example, CEVAE [24] uses variational autoencoders (VAE) to estimate a latent-variable model that simultaneously discovers hidden confounders and infers how they affect treatment and outcome. Another example is GANITE [43], which uses generative adversarial networks (GAN) to estimate treatment effects. This approach attempts to capture the uncertainty in the counterfactual distributions and is defined for any number of treatments.

Several studies, such as [28] and [42], have investigated the process of modeling all Independent Causal Mechanisms (ICMs) to derive a Structural Causal Model (SCM). This comprehensive method carries its own challenges and advantages, making it a suitable option for certain scenarios. The methodology presented in this paper finds a balance between fully connected modeling and SCM-based modeling.

A careful inspection of the methods shows that the use of new algorithmic inductive biases that encode prior information about the causal structure of data has been one of the most successful strategies to build better causal effect estimation methods [21]. Typically, these approaches fall into one of two categories: altering the network's structure, exemplified by the branching found in TARNet, or modifying the loss function, as seen with the positivity robust loss utilized in BCAUSS.

In Table 2 some of the most relevant causal inference methods are considered along with the biases they address. It shows which methods can correct or mitigate each type of bias. Using data efficiently and ensuring treatment relevance are among the most critical sources of error and have been addressed by most models. We can also see that one of the most interesting properties of neural network models is that they can stack multiple correction strategies to address several sources of bias at the same time.

III. A NEW INDUCTIVE BIAS FOR ESTIMATING CAUSAL EFFECTS

A. SPURIOUS VARIABLE INTERACTIONS AND EFFECTS ESTIMATION

A **variable interaction** refers to a situation in which the effect of one variable on an outcome depends on the value of another variable. In other words, the relationship between two or more variables is not purely additive or independent; instead, the variables influence each other in ways that modify their combined impact on the outcome.

For continuous variables, interactions can be mathematically characterized using cross-partial derivatives of the outcome function. Consider a function $f_Y(x_1, x_2)$ representing

TABLE 2. Comparison of some relevant causal inference methods in terms of the biases they address. Checkmarks indicate that the method addresses the bias indicated in the column. In this paper, we present SIP, a method that can be applied on top of other representation-based learners to exploit the knowledge about the ICM structure of the settings. With a solid arrow, we represent the methods evaluated. With a dashed arrow, other representation-based methods where SIP could be applied.

Method	Identification Bias				Estimation Bias			
	Exchangeability	Positivity	No interference	Consistency	Data efficiency	Treatment relevance	Propensity	Spurious Interactions
Linear Models [40]	✗	✗	✗	✗	✓	✗	✗	✓
S-learner [21]	✗	✗	✗	✗	✓	✗	✗	✗
T-learner [21]	✗	✗	✗	✗	✗	✓	✗	✗
X-learner [21]	✗	✗	✗	✗	✓	✓	✗	✗
B-Learner [28]	✓	✗	✗	✗	✓	✓	✓	✗
TARNet[36]	✗	✗	✗	✗	✓	✓	✗	+ SIP → ✓
CFRNet [36]	✗	✗	✗	✗	✓	✓	✓	+ SIP → ✓?
Dragonnet [37]	✗	✗	✗	✗	✓	✓	✓	+ SIP → ✓
BCAUSS [39]	✗	✓	✗	✗	✓	✓	✓	+ SIP → ✓
HINITE [23]	✗	✗	✓	✗	✓	✓	✓	+ SIP → ✓?

the relationship between two variables x_1 , x_2 , and an outcome Y . The presence of an interaction between x_1 and x_2 can be assessed by evaluating the second mixed derivative:

$$\frac{\partial^2 f_Y(x_1, x_2)}{\partial x_1 \partial x_2}. \quad (3)$$

If this cross-partial derivative is nonzero, it indicates that the effect of x_1 on Y depends on the value of x_2 , and vice versa, signifying an interaction. Conversely, if the derivative is zero, the variables do not interact, and their effects on the outcome are independent of one another. An analogous expression involving finite differences can be used for categorical variables and, for machine learning models, interactions can be characterized using sampling-based methods [12].

Consider an observational scenario where Y is the target variable, T is the treatment, X represents a valid adjustment set, and $f_Y^T(X)$ is an estimated function that models the causal effect as identified in Equation 2. It is important to be aware that X generally includes variables that are not strictly necessary according to the backdoor criterion.¹ These variables, referred to as **neutral controls** in the literature, can reduce the variance of the estimate without affecting its bias [3].

The consideration of neutral controls in X is interesting to increase the precision of the estimand but introduces a new kind of bias: the potential reliance of $f_Y^T(X)$ on variable interactions that are not consistent with the data generation process. The ICM principle asserts that the data-generating process is composed of a set of independent causal mechanisms. However, the inclusion of neutral controls in the adjustment set can result in the model $f_Y^T(X)$ failing to preserve this independence.

Consider the causal graph in Fig. 1, where x_4 is a neutral control. In this case, if $f_Y^T(X)$ relies on the interaction between x_2 and x_4 , ($\frac{\partial^2 f_Y^T(X)}{\partial x_2 \partial x_4} \neq 0$), it will use an interaction not present in any ICM. We refer to these types of variable interactions as **spurious interactions**.

¹ See details about how to build the adjustment sets in Appendix A.

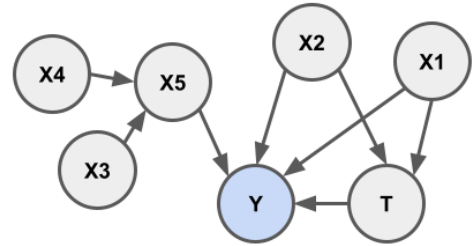


FIGURE 1. This example causal graph can be factorized as follows: $P(Y, T, x_1, x_2, x_3, x_4, x_5) = P(Y|T, x_1, x_2, x_5)P(T|x_1, x_2)P(x_5|x_3, x_4)P(x_1)P(x_2)P(x_3)P(x_4)$. Note, for example, that x_4 and x_2 do not appear together in any single term.

B. IMPACT OF SPURIOUS INTERACTIONS IN CAUSAL EFFECT ESTIMATION

To estimate causal effects, it is necessary to approximate the distribution of the data under intervention, which gives rise to the challenges described in Section II.

Among the possible sources of associations in the data, only those arising from causal relationships reflect the intrinsic dependencies between variables [4]. Other types of associations are spurious and depend on the joint distribution of features and data collection processes, making them sensitive to changes in these factors. Spurious interactions, which fail to preserve the independence structure inherent to Independent Causal Mechanisms (ICMs), do not arise from causal relationships between variables. Consequently, they can compromise the model's robustness to interventions.

Estimand models are designed to learn linear or nonlinear relationships between input and output variables. The highest-performing models for a given training dataset often leverage subtle statistical associations among features. However, these associations may not stem from causal relationships, making the model prone to prediction errors under interventions and compromising its reliability for estimating causal effects. Returning to the example in Fig. 1, using an interaction between x_2 and x_4 in a model would correspond to modeling a different causal graph—one where, for instance, x_4 is a direct cause of Y .

The issue of bias arising from variable interactions has been explored in the context of linear models [13], [40], where each interaction's inclusion must be carefully evaluated, and in tree-based predictive models [11], [33]. However, this topic has received little attention in the area of neural networks, despite their reliance on interactions among all variables.

C. FUNCTIONAL FORM OF MODELS WITHOUT SPURIOUS INTERACTIONS

To address the issue of nonlinear models relying on spurious interactions, we propose introducing a novel modeling principle for $f_Y^T(X)$, aimed at improving the robustness and causal validity of treatment effect estimation. This principle explicitly prioritizes interactions rooted in the true generative process while excluding those that are spurious or irrelevant. **Spurious Interactions Principle:** *To estimate a robust model of causal effects, the model should rely solely on variable interactions present in the Independent Causal Mechanisms (ICMs) of the generative process, disregarding any spurious interactions.*

This principle fundamentally aligns the model with the causal structure underlying the data. Specifically, it restricts $f_Y^T(X)$ to only consider variable interactions that are consistent with the causal mechanisms responsible for generating the observed data. By doing so, the model avoids overfitting spurious patterns that do not reflect genuine causal relationships.

Formally, given a set of K Independent Causal Mechanisms (ICMs) in the data generating process, let G_k denote the subset of variables involved in the k -th ICM. Under the Spurious Interactions Principle, the model $f_Y^T(X)$ must adhere to the following constraint:

$$\frac{\partial^2 f_Y^T(X)}{\partial x_i \partial x_j} \neq 0 \iff (i, j) \in G_k, \forall k \quad (4)$$

This constraint ensures that the model's representation of variable interactions is aligned with the causal structure of the data. By focusing exclusively on interactions prescribed by the ICMs, the model avoids introducing dependencies that are inconsistent with the generative process. This approach not only enhances the causal interpretability of the model but also reduces the risk of overfitting to spurious correlations present in the training data.

The Spurious Interactions Principle is particularly relevant for neural network models, which are often predisposed to exploit arbitrary variable interactions due to their high flexibility. By embedding causal constraints directly into the modeling process, our principle introduces a safeguard against this vulnerability, ensuring that the learned representations remain faithful to the underlying causal mechanisms. This alignment is crucial for achieving robust and generalizable treatment effect estimations, particularly in settings where spurious patterns can lead to misleading inferences.

In Section IV, we delve into the methodology for integrating this inductive bias into neural network models to enhance their robustness and causal fidelity. We provide a detailed explanation of how the Spurious Interactions Principle can be

operationalized within the architecture and training process of neural networks. Specifically, we discuss the strategies for constraining the network's reliance on variable interactions, ensuring that these interactions align with the Independent Causal Mechanisms (ICMs) defined by the generative process.

IV. SPURIOUS INTERACTIONS PREVENTION

In this section, we propose a comprehensive methodology for incorporating a new inductive bias that leverages the Spurious Interactions Principle in neural network models. This approach can utilize expert insights on the relationships between specific variables or knowledge about the causal graph and can be seamlessly integrated with most existing inductive biases.

A. HOW TO BUILD THE INDUCTIVE BIAS INTO NEURAL NETWORK MODELS

To apply this principle, and considering that most state-of-the-art methods for estimating causal effects are based on representation learning, we propose implementing a new inductive bias in the pre-representation stage of these models. This inductive bias must ensure that the generated representation Z , which encodes information relevant to the computation of causal effects, does not rely on spurious interactions and is compatible with the inductive biases of other methods.

Let $f_k(G_k)$ denote a (non-linear) function of the subset of variables participating in the k -th ICM of the data generating process, ICM_k . To construct a multidimensional representation Z of X that does not rely on spurious interactions, **each feature z_j of Z can only be expressed as a linear combination of the features of some $f_k(G_k)$** . Any other combination would create interactions between variables from different ICMs, resulting in spurious interactions that are not robust to interventions and would consequently reduce the model's performance in estimating causal effects.

Hence, the only form of the causal effects model that can encode the inductive bias we propose is a representation learning model with a pre-representation stage $f_Z(X)$ and a post-representation stage $f_Y(f_Z(X))$ such that:

$$f_Z(X) = [f_1(G_1) \parallel f_2(G_2) \parallel \dots \parallel f_K(G_K)] \cdot A, \quad (5)$$

where every f_k is a non-linear function, possibly modelled with a neural network, whose outputs are concatenated and A is a learned weights matrix, possibly implemented as a fully connected layer in a neural network.

It is interesting to note that any interaction from distinct ICMs is spurious. This includes different ICMs with the same set of variables. This concept is akin to that in linear modeling, where a single variable may be included several times under various transformations, such as squaring or taking the logarithm. In section 4.2.1 we perform experiments that test this scenario and in the Appendix B we explore it using the illustrative example.

The selection of which groups G_i to include in the representation Z is done graphically by selecting the

ICMs related to the minimal feature subset with maximum predictivity [44]. In the backdoor setting, this includes the ICMs for the target variable Y and for each of its parents $Pa(Y)$. This criteria can be expanded or modified with expert knowledge about the ICMs. Additionally, the variable T is excluded from all the groups because the learners implemented in this paper do not use T as an input but rather as part of the structure. You can see an example in Figure 2.

A direct consequence of using the proposed inductive bias is that all valid interactions are included in the representation Z , while all discarded interactions are of spurious nature.

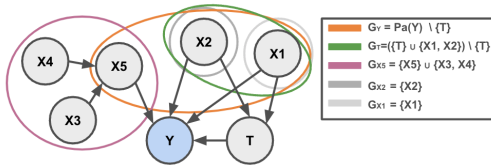


FIGURE 2. Example of groups of variables in the ICMs related to the target Y . In this setting, the groups of variables defined by the encoded ICM are: $\{x_1, x_2\}$, $\{x_1, x_2, x_5\}$, $\{x_3, x_4, x_5\}$, $\{x_1\}$, $\{x_2\}$.

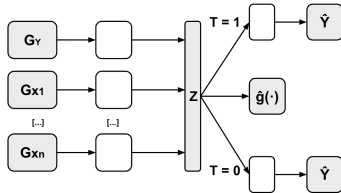


FIGURE 3. Model architecture when applying SIP to the Dragonnet. Post-representation remains identical but pre-representation layers are divided according to the groups of variables.

B. LIMITATIONS

In principle, determining spurious interaction involves, either in part or completely, knowledge of the data generating process. In the case of not having access to the causal graph, we can approximate $\{G_k\}$ by using expert knowledge about existing ICMs or by considering causal discovery methods [38]. This approach helps limit our model's reliance on at least some of the non-robust relationships between covariates and the target variable. In Appendix B, we analyze the performance of the method under various levels of causal information quality and show that our method is robust to imperfect causal graphs and that using partial causal information is preferable to ignoring it.

C. IMPLEMENTATION AND NEURAL NETWORK ARCHITECTURE

Similarly to other methods related to representation learning field, our technique can be broken down into two stages. Initially, we define the inputs and the layers that lead to the representation. Subsequently, we analyze the representation layer and the output layers of the network. A pseudocode for the implementation can be found in Algorithm 1 and a visualization of the model in Pytorch [30] in appendix D.

1) PRE-REPRESENTATION: VARIABLE GROUPS

The input of the network is divided into different groups of variables, as shown in Figure 3. We create a set of layers for each variable group G_k that do not interact with each other. Each set of layers receives only the variables in the group as input. Note that a single variable can be part of more than one group. The output of all these layers is concatenated as an input to the linear merge layer that constitutes a representation Z . This ensures that the representation encoded in Z , which is used as input for the rest of the network, is free from spurious interactions.

2) POST-REPRESENTATION: OUTPUT HEADS

Most state of the art methods modify the output of the network by changing the architecture after the representation layer, i.e. considering distinct heads or loss functions. In contrast, our methodology takes place in the architecture before the representation layer. This allows for a combination of both architectures, using our methodology combined with any of the output heads. In this paper we implemented the TARNet [36], Dragonnet [37] and BCAUSS [39] heads, but other representation-based learners can be connected to the network.

Algorithm 1 SIP Procedure

Input: Train, test sets $(D_{tr}, D_{te}) \sim D$ with treatment T , target Y and valid adjustment set X .

Parameter: A representation-based $f_Y^T(f_Z(X))$ estimator based on a neural network NN .

- 1: Causal Knowledge Specification: List all forbidden variable interactions. Build (or discover) the associated causal graph or use expert knowledge.
- 2: Group Determination: Create the groups of variables $G_{x_i} := (\{x_i\} \cup An(x_i)) \setminus \{T\}$, $\forall x_i \in Pa(Y)$ and $G_Y := Pa(Y) \setminus \{T\}$. If the causal graph is not available, form all compatible subsets of variables.
- 3: Architecture (Pre-representation): Create an independent set of layers for each group, all linearly connected to the representation layer Z .
- 4: Architecture (Post-representation): Use the head of your choice (TARNet, Dragonnet, BCAUSS, etc.) from the representation layer onward.
- 5: Model: Train the NN using D_{tr} and D_{te} and run the causal inference workflow.

V. EMPIRICAL TESTING

Our objective is to assess whether models restricted by the proposed method achieve superior performance compared to their unrestricted counterparts. The suggested method is versatile and can be applied to a wide range of representation-based learners. For this evaluation, we focus on three of the most effective models in causal inference: TARNet [36], Dragonnet [37], and BCAUSS [39]. By incorporating both SIP-constrained and unconstrained variants for each model, we create a total of six distinct models for experimentation.

This setup allows us to conduct a comprehensive analysis of the method's impact on performance. Specifically, we aim to determine whether the imposition of SIP enhances the models' ability to improve causal effect estimation. By leveraging a diverse set of state-of-the-art learners, we ensure that the findings are robust and applicable across different architectures and scenarios.

We break down our research questions into how relevant the prevention of spurious interactions is, and how well the method works under practical scenarios.

- **RQ1.** What is the impact of spurious interaction bias in state-of-the-art causal estimation methods? Does it retain significance when other estimation biases are taken into account?
- **RQ2.** What is the impact of having imperfect or partial causal knowledge when addressing spurious interactions?
- **RQ3.** How well does the method perform in practical real world scenarios?

We have designed an experiment using a synthetic data benchmark to address the first two research questions. This involves a series of synthetic datasets introduced in [26] and implemented in [2]. These experiments encompass four scenarios (A–D) of varying complexity (see subsection V-B1). For RQ1 we use the knowledge of the data generating process to prevent all spurious interactions. For RQ2, we employ causal discovery techniques, as outlined in Section V-A, to identify and classify interactions into valid and spurious groups. In addition to the synthetic datasets, we also incorporate semi-synthetic data, the IHDP dataset [16], to further evaluate the approach. Finally, for RQ3, we evaluate our approach using the Jobs real-world dataset [22].

A. EXPERIMENTAL SETUP

Evaluating causal inference methods is hard because we usually have no access to counterfactual data. To overcome this problem, one of the common approaches is to use semi-synthetic data. We test the models on two of the most common benchmarks in causal inference: IHDP and Jobs. These benchmarks have already been used by some of the best performing methods and thus are specially relevant for the comparison of SIP-constrained and unconstrained models.

Since these benchmarks might present some problems [5], we also test the models on a range of synthetic benchmarks that represent distinct situations, details in the synthetic experiments section. All the code for reproducing the experiments can be found in <https://anonymous.4open.science/r/NNSIP-4DB7>. All the experiments are run on a MacBook Pro with a Apple M2 Max chip and 32GB memory. Hyperparameters and evaluation metrics used to run the experiments can be found in the code or in the section C of the appendix. In this work, we assume all our datasets are originally composed of valid adjustment sets, i.e. the backdoor criterion can be satisfied.

1) CAUSAL GRAPHS

Causal discovery in the experiments is performed by using the IcaLinGam algorithm [38] implemented in [45]. The output of this algorithm usually involves only a subset of the variables. Since we are interested in the effect of using correct information rather than the quality of the discovery, we group the remaining variables into a single group. Errors in the discovery might still translate to loss of performance of the constrained models. In appendix B we perform an exploration of how the precision of the discovery might affect the model accuracy. The choice of discovery algorithm is out of the scope of this work, and IcaLinGam has been selected for its ability to converge under distinct scenarios. Given that the adjustment set is considered valid, then all nodes linked to the target must be in the parents of Y to comply with the backdoor criterion.

B. DESCRIPTION OF DATASETS USED

1) SYNTHETIC EXPERIMENTS

We use a set of synthetic experiments defined in [26] and implemented in [2]. These experiments are composed of four scenarios: scenario A is a complex outcome regression model with an easy treatment effect; scenario B is a randomized controlled trial; scenario C has an easy propensity score with a difficult control outcome; and scenario D has an unrelated treatment arm and control arm. A common characteristic of all scenarios is that they each possess an ICM that encompasses all covariates. This implies that any variable interactions are allowed at least on one ICM, and SIP's role is solely to inhibit interactions among separate ICMs. We run each scenario using the settings in the original paper: number of samples $n \in \{500, 1000\}$, number of variables $d \in \{6, 12\}$, and noise level $\sigma \in \{0.5, 1, 2, 4\}$. We generate an equal number of train and test samples. Each configuration is repeated 100 times, making a total of 6400 runs. We average for the number of samples and number of variables for controlling the reporting size and show the results at the scenario-noise level for its relevance.

2) SEMISYNTHETIC EXPERIMENTS: IHDP

The Infant Health and Development Program (IHDP) is a study that used randomization to examine the effects of home visits by specialist doctors on the cognitive test scores of premature infants. The dataset² was first used by [16] to evaluate algorithms for estimating treatment effects. To create an observational dataset, non-random subsets of treated individuals were removed, resulting in selection bias. The outcomes in the dataset were generated using the original covariates and treatments. The dataset consists of 747 subjects and 25 variables. Following the recent literature, we used the simulated outcome implemented in the NPCI package [8], which is composed of 1000 repetitions of the experiment. We averaged our results over 1000 train/validation/test splits with ratios 70/20/10.

²The dataset can be accessed at www.fredjod.com

TABLE 3. Ratio between constrained and unconstrained models' PEHE (left) and ATE (right) on the synthetic benchmark with known spurious interactions. Values < 1 mean the constrained version has a lower error. All reported values have a standard error of the mean < 0.025.

scenario	noise	TARNet	Dragonnet	BCAUSS
A	1/2	0.74	0.72	0.94
	1	0.89	0.86	1.01
	2	1.09	1.12	1.03
	4	1.19	1.23	1.01
B	1/2	0.66	0.70	0.56
	1	0.88	0.90	0.73
	2	1.01	1.07	0.87
	4	1.13	1.08	0.94
C	1/2	0.47	0.46	0.35
	1	0.70	0.76	0.65
	2	0.90	0.91	1.07
	4	1.19	1.10	1.10
D	1/2	0.37	0.38	0.30
	1	0.56	0.58	0.59
	2	0.78	0.78	0.89
	4	0.98	0.97	1.00
scenario	noise	TARNet	Dragonnet	BCAUSS
A	1/2	0.68	0.66	0.95
	1	0.77	0.75	1.02
	2	0.95	0.95	1.03
	4	1.06	1.14	1.02
B	1/2	0.82	0.88	0.70
	1	0.92	0.96	0.74
	2	0.91	0.99	0.84
	4	1.05	0.94	0.90
C	1/2	0.46	0.44	0.16
	1	0.48	0.51	0.28
	2	0.62	0.61	0.54
	4	0.84	0.78	0.74
D	1/2	0.80	0.92	0.62
	1	0.98	1.02	0.79
	2	1.02	1.02	0.99
	4	1.05	1.02	1.06

3) REAL DATA EXPERIMENTS: JOBS

The LaLonde Jobs dataset² [22] assesses the impact of job training as a treatment on income and employment status after training. We use the feature set defined in [7]. Following [36], we combined the LaLonde experimental sample (297 treated, 425 control) with the PSID comparison group (2490 control). We averaged over 10 train/validation/test splits with a ratio of 62/18/20. Due to the small size of the benchmark we repeat the process 10 times and report the average of the results. For most of the runs, the discovery algorithm was unable to find a causal graph, and a default was used. The default causal graph was obtained as the most common causal graph discovered across 100 runs.

C. EXPERIMENT RESULTS

1) ANSWERING RQ1: SYNTHETIC EXPERIMENTS

The results of the synthetic experiments are shown in Table 3 and Table 4. We observe that constrained

TABLE 4. Test metrics on the synthetic benchmark with known spurious interactions averaged to the scenario level. Lower is better. All reported values have a standard error of the mean < 0.025. T., D., B. correspond to TARNet, Dragonnet and BCAUSS respectively.

Model	$\sqrt{\epsilon_{PEHE}}$				ϵ_{ATE}			
	A	B	C	D	A	B	C	D
T.	.355	.342	.438	.570	.281	.189	.322	.165
D.	.355	.342	.442	.569	.280	.193	.329	.161
B.	.410	.436	.429	.682	.341	.220	.412	.166
T. + SIP	.350	.337	.387	.414	.244	.184	.216	.166
D. + SIP	.354	.340	.384	.413	.245	.183	.214	.163
B. + SIP	.409	.349	.362	.481	.342	.185	.200	.157

models generally perform better than their unconstrained counterparts.

Among the models incorporating SIP, TARNet + SIP achieves the lowest PEHE values in three out of four scenarios, while BCAUSS + SIP delivers the lowest ATE values in three out of four scenarios. Dragonnet + SIP performs similar to TARNet + SIP and surpasses the performance of the unconstrained models.

In scenarios with higher variance—characterized by noise levels of 4 for ATEs and 2 or 4 for PEHEs—the unconstrained models performed similarly to or better than their constrained counterparts. This indicates that in high-noise environments, modeling specific mechanisms becomes more challenging, potentially limiting the effectiveness of the SIP constraint.

Each of the selected methods addresses one or more estimation biases (see Section II-A). However, the experiments reveal that the bias arising from spurious variable interactions is not mitigated by these methods. In fact, addressing this specific bias can, in some cases, significantly improve the accuracy of the estimator.

2) ANSWERING RQ2: SYNTHETIC AND SEMI-SYNTHETIC EXPERIMENTS

Tables 5 and 6 present the results of the synthetic benchmark when the data-generating process is not known, using the same format as in the previous section. While the performance is slightly lower compared to the fully known case, the constrained models consistently outperform their unconstrained counterparts in most scenarios. Notably, all top-performing models fall within the constrained category.

We have also examined the IHDP dataset, which consists of semi-synthetic data (see subsection V-B2). As shown in Table 7, the constrained BCAUSS model achieves the best performance on both the ate and pehe metrics, with its unconstrained counterpart ranking second.

The pehe and ate metric values of the unconstrained models (TARNet, Dragonnet, and BCAUSS) were higher than those of the constrained models on both the train and test sets, indicating that the constraints reduced the estimation error and increased the generalization ability of the models. These results indicate that partial causal information is sufficient to

TABLE 5. Ratio between constrained and unconstrained models' PEHE (left) and ATE (right) on the synthetic benchmark with discovered spurious interactions. Values < 1 mean the constrained version has a lower error. All reported values have a standard error of the mean < 0.025 .

scenario	noise	TARNet	Dragonnet	BCAUSS
A	1/2	0.76	0.76	0.93
	1	0.91	0.92	1.02
	2	1.07	1.04	1.06
	4	1.08	1.11	1.02
B	1/2	0.73	0.72	0.63
	1	0.92	0.94	0.78
	2	1.02	1.06	0.94
	4	1.00	0.99	0.96
C	1/2	0.51	0.51	0.36
	1	0.71	0.76	0.70
	2	1.00	1.00	1.09
	4	1.12	1.13	1.18
D	1/2	0.5	0.51	0.38
	1	0.76	0.76	0.72
	2	0.92	0.92	0.93
	4	1.00	1.00	1.01
scenario	noise	TARNet	Dragonnet	BCAUSS
A	1/2	0.73	0.74	0.93
	1	0.83	0.81	1.02
	2	0.96	0.92	1.07
	4	1.01	1.04	1.02
B	1/2	0.87	0.88	0.69
	1	0.96	0.98	0.87
	2	0.95	0.99	0.98
	4	0.91	0.93	0.95
C	1/2	0.58	0.58	0.17
	1	0.54	0.64	0.31
	2	0.70	0.72	0.54
	4	0.87	0.91	0.79
D	1/2	0.85	0.92	0.7
	1	0.98	1.01	0.88
	2	1.01	0.99	0.94
	4	1.02	1.03	1.07

TABLE 6. Test metrics on the synthetic benchmark with discovered spurious interactions averaged to the scenario level. Lower is better. All reported values have a standard error of the mean < 0.025 . T., D., B. correspond to TARNet, Dragonnet and BCAUSS respectively.

Model	$\sqrt{\epsilon_{PEHE}}$				ϵ_{ATE}			
	A	B	C	D	A	B	C	D
T.	.363	.350	.456	.564	.285	.197	.316	.158
D.	.359	.349	.446	.568	.283	.196	.306	.160
B.	.413	.432	.424	.681	.346	.212	.405	.163
T. + SIP	.350	.334	.411	.475	.252	.182	.232	.158
D. + SIP	.347	.335	.408	.478	.250	.185	.238	.161
B. + SIP	.417	.366	.378	.523	.348	.195	.207	.157

enhance the performance of the proposed method. However, its effectiveness is contingent on the quality of the provided causal information. For further analysis of this scenario, refer to Appendix B.

TABLE 7. Train (top) and test (bottom) metrics on the IHDP benchmark. The "Comp" value is the ratio between the constrained and the unconstrained model.

Model	$\sqrt{\epsilon_{PEHE}}$		ϵ_{ATE}	
	Value	Comp	Value	Comp
BNN [18]	$2.2 \pm .1$	-	$.37 \pm .03$	-
CFRW [36]	$.71 \pm .0$	-	$.25 \pm .01$	-
CEVAEs [24]	$2.7 \pm .1$	-	$.34 \pm .01$	-
GANITE [43]	$1.9 \pm .4$	-	$.43 \pm .05$	-
TARNet + SIP	$.960 \pm .032$	0.78	$.148 \pm .004$	1.04
TARNet	$1.23 \pm .046$		$.142 \pm .004$	
Dragonnet + SIP	$.974 \pm .034$	0.797	$.138 \pm .003$	.968
Dragonnet	$1.22 \pm .047$		$.143 \pm .004$	
BCAUSS + SIP	$.663 \pm .022$	.805	$.100 \pm .003$	.929
BCAUSS	$.824 \pm .023$		$.107 \pm .003$	

Model	$\sqrt{\epsilon_{PEHE}}$		ϵ_{ATE}	
	Value	Comp	Value	Comp
BNN [18]	$2.1 \pm .1$	-	$.42 \pm .03$	-
CFRW [36]	$.76 \pm .0$	-	$.27 \pm .01$	-
CEVAEs [24]	$2.6 \pm .1$	-	$.46 \pm .02$	-
GANITE [43]	$2.4 \pm .4$	-	$.49 \pm .05$	-
TARNet + SIP	$1.014 \pm .039$	0.793	$.184 \pm .006$	0.911
TARNet	$1.279 \pm .052$		$.201 \pm .008$	
Dragonnet + SIP	$1.030 \pm .044$	0.797	$.188 \pm .009$	0.895
Dragonnet	$1.291 \pm .057$		$.204 \pm .010$	
BCAUSS + SIP	$.749 \pm .031$	0.777	$.131 \pm .005$	0.864
BCAUSS	$.963 \pm .034$		$.151 \pm .006$	

TABLE 8. Train and test metrics on the Jobs benchmark. Lower is better. The "Comp" value is the ratio between the constrained and the unconstrained model.

Model	ϵ_{ATT}^{tr}		ϵ_{ATT}^{te}	
	Value	Comp	Value	Comp
BNN [18]	$.04 \pm .01$	-	$.09 \pm .04$	-
CFRW [36]	$.04 \pm .01$	-	$.09 \pm .03$	-
CEVAEs [24]	$.02 \pm .01$	-	$.03 \pm .01$	-
GANITE [43]	$.01 \pm .01$	-	$.06 \pm .03$	-
TARNet + SIP	$.037 \pm .003$	0.945	$.087 \pm .008$	1.022
TARNet	$.039 \pm .004$		$.085 \pm .008$	
Dragonnet + SIP	$.054 \pm .010$	1.121	$.097 \pm .012$	1.147
Dragonnet	$.048 \pm .004$		$.085 \pm .008$	
BCAUSS + SIP	$.017 \pm .006$	0.288	$.075 \pm .007$	0.796
BCAUSS	$.058 \pm .001$		$.094 \pm .009$	

3) ANSWERING RQ3: REAL WORLD SCENARIO

Jobs is a small benchmark and the performance of the models varies greatly between runs. This characteristic is evidenced in the standard deviation values, which are in the same order as the mean in most cases in Table 8. Because of that, we think the use of the benchmark rests upon its ability to test that models perform well on real data. In this sense, all models perform comparably well. The lowest ATT error for train is obtained by GANITE followed by the constrained version of BCAUSS while for test, the lowest error is obtained by CEVAE followed by GANITE. In this dataset, the discovery of the causal graph is especially unreliable, and this is reflected in the comparison

between constrained and unconstrained models, which obtain similar results.

VI. DISCUSSION AND CONCLUSION

We have presented SIP, a novel method for incorporating causal information into the estimation of heterogeneous treatment effects. Our method leverages an inductive bias that exploits the ICM principle in neural network models and can be applied on top of other representation learning methods. We have tested the effectiveness of our method using three different base models. The experiments indicate that constraining models using the described method leads to significant improvements, achieving new state-of-the-art results.

SIP is flexible and performed well for all three base models. Following the idea of stacking mechanisms to address sources of error, SIP can be combined with other techniques to introduce additional inductive biases. We have also shown that our method is robust to imperfect causal graphs, and that using partial causal information is preferable to ignoring it. Having no restrictions on interactions is the same as assuming a causal graph that is fully connected, and in most cases there are at least some known edges that are not causal.

As a future work, we plan to enhance our method by using masking and a graphical conditioner to build a neural network that prevents spurious interactions by design, sharing weights among the groups. This would reduce redundancy in the architecture and make a more efficient use of the data.

APPENDIX A ADJUSTMENT SETS

Even though the backdoor criterion provides a complete identification of all **valid adjustment sets** (sets of variables can be used to identify the causal effects, possibly including variables that are not strictly necessary), it does not tell us which is the **optimal adjustment set** for practical estimation in finite-sample scenarios.

Those variables in a valid adjustment set that are not strictly required for identification are called **neutral controls** [3]. These variables neither increase nor decrease the asymptotic bias of the estimation of treatment effects, but they can still have a huge impact in a finite-sample scenario and can improve the performance of estimators. The optimal adjustment set usually contains neutral controls.

There are advanced methods that, given an scm, allow to define **efficient adjustment sets** (adjustment sets that yield the least squares estimator with the smallest asymptotic variance) [15], [34]. Although this scenario is not addressed in our work, it's important to note that these sets may also be prone to spurious interaction bias. Consequently, the method outlined in this paper is also applicable to them.

APPENDIX B ILLUSTRATIVE EXAMPLE

We create an illustrative example that follows the causal graph in Fig. 2 to have full control over the scenarios generated. The

data generating process is as follows:

$$p = 4$$

$$X \sim \text{Uniform}(0, 1)^{n \times p}$$

$$\eta = 0.1$$

$$e = \text{Logistic}(\max(\eta, \min(\sin(\pi X_{:,1} X_{:,2}), 1 - \eta)))$$

$$e2 = \text{Logistic}(\max(\eta, \min(\sin(\pi X_{:,3} + X_{:,4}), 1 - \eta))) + \text{Uniform}(0, 1)^n$$

$$\tau \sim \mathcal{N}(1, 1)^n$$

$$w \sim \text{Binomial}(1, e)^n$$

$$y = 2 \times \log(1 + \exp(X_{:,1} + X_{:,2} + (w - 0.5) \times \tau + e2)) + \mathcal{N}(0, 1)^n$$

We create the following scenarios to represent distinct knowledge about the data generating process. All results are obtained using the Dragonnet as base learner.

- Scenario 1 - Dragonnet.
- Scenario 2 - A single ICM with all variables
- Scenario 3 - No neutral controls ICM, just G_Y and G_T .
- Scenario 4 - No individual variables ICM: G_Y , G_T , G_{X5} .
- Scenario 5 - Causal graph available.

In this illustrative example results in table 9, model performance increases progressively as more information about the causal structure is encoded in SIP.

TABLE 9. Results for PEHE and ATE across different scenarios of the illustrative example. Values represent the ratio between the scenario and the baseline model.

Scenario	PEHE	ATE
1 - Baseline	1.00	1.00
2	0.96	0.88
3	0.94	0.85
4	0.94	0.83
5	0.93	0.79

A. CAUSAL DISCOVERY PERFORMANCE

We generate data with increasing noise levels and test the performance of Dragonnet+SIP with a discovered causal graph. We test the accuracy of causal discovery for the synthetic datasets in section 4.1. For each scenario the causal graph is discovered and compared to the real causal graph using the Graph Edit Distance (GED). We then aggregate at noise level and compare GED with model performance in table 10.

TABLE 10. Results for GED, Y_noise, PEHE, and ATE.

Y_noise	GED	PEHE	ATE
0.5	2.88	0.60	0.73
1.0	3.14	0.79	0.79
2.0	4.06	0.99	0.85
4.0	5.55	1.05	0.93

This results suggest that noise present in the data decrease the accuracy of causal discovery and this in turn affects the performance of SIP.

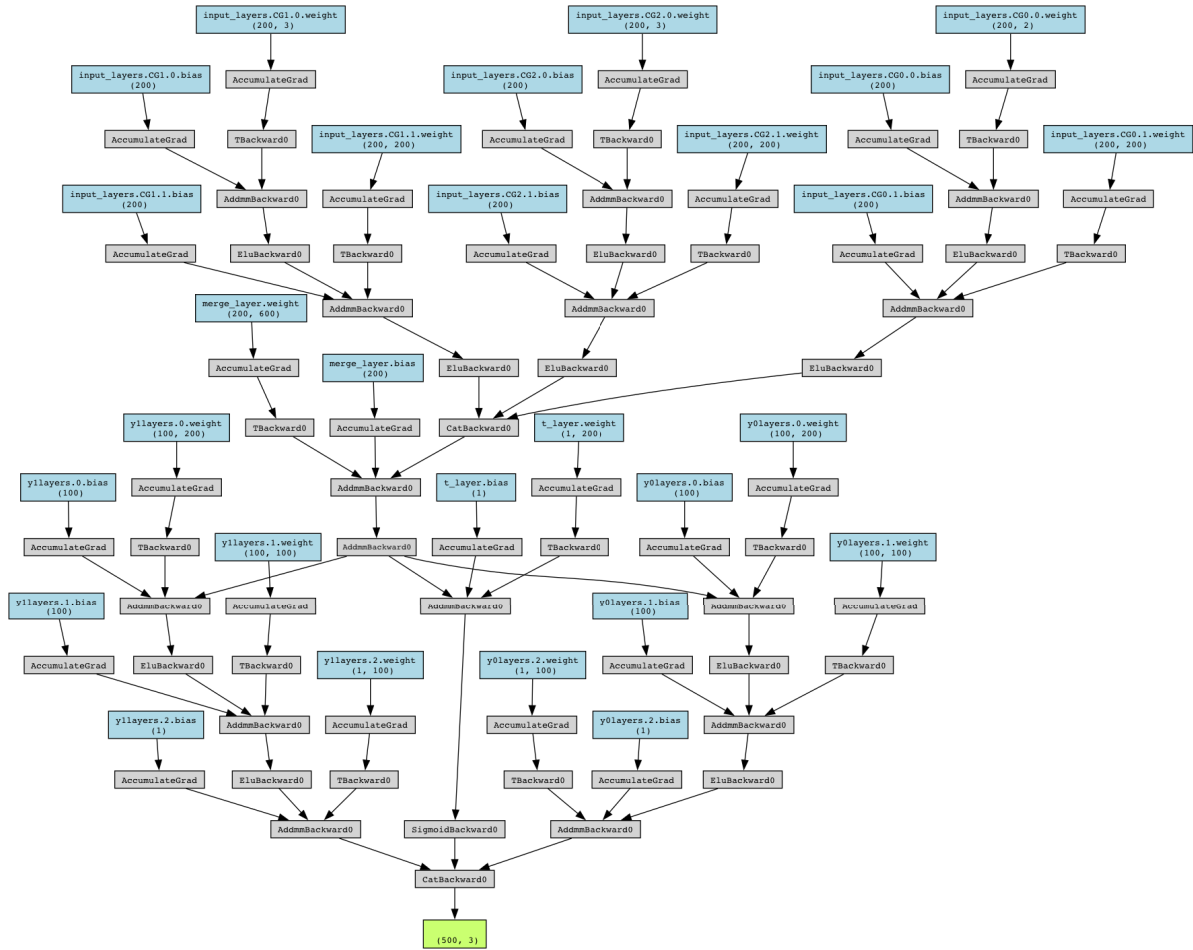


FIGURE 4. Example of Dragonnet + SIP implemented in Pytorch.

APPENDIX C HYPERPARAMETERS AND EVALUATION METRICS

B. HYPERPARAMETERS

We use the same training hyperparameters defined in [37] and [39], as they show to achieve state of the art results. These include a hidden layer size of 200 for the shared representation layers and 100 for the conditional outcome layers, train using stochastic gradient descent with learning rate 10^{-5} and momentum 0.9, and batch size equal to 64 for the TARNet and Dragonnet and equal to the train-set length for BCAUSS. Details can be found in the code implementation.

In the constrained architectures, the hyperparameters for each set of layers are kept the same as those in the unconstrained architectures. However, the activation function of the representation layer is modified to a linear function.

C. EVALUATION METRICS

When counterfactual data is accessible, we use standard metrics such as the error in Average Treatment Effects -ate- and the Precision in Estimation of Heterogeneous Effects -pehe-. For the Jobs dataset, no counterfactuals are available, so we use the experimental nature of the data to compute the error of the average treatment effects on the treated att.

APPENDIX D SIP ARCHITECTURE EXAMPLE WITH PYTORCH

See Figure. 4.

REFERENCES

- [1] Y. Bengio, A. Courville, and P. Vincent, "Representation learning: A review and new perspectives," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 8, pp. 1798–1828, Aug. 2013.
- [2] H. Chen, T. Harinen, J.-Y. Lee, M. Yung, and Z. Zhao, "CausalML: Python package for causal machine learning," 2020, *arXiv:2002.11631*.
- [3] C. Cinelli, A. Forney, and J. Pearl, "A crash course in good and bad controls," *Sociol. Methods Res.*, vol. 53, no. 3, pp. 1071–1104, Aug. 2024.
- [4] P. Cui and S. Athey, "Stable learning establishes some common ground between causal inference and machine learning," *Nature Mach. Intell.*, vol. 4, no. 2, pp. 110–115, Feb. 2022.
- [5] A. Curth, D. Svensson, J. Weatherall, and M. V. D. Schaar, "Really doing great at estimating CATE? A critical look at ML benchmarking practices in treatment effect estimation," in *Proc. Thirty-fifth Conf. Neural Inf. Process. Syst. Datasets Benchmarks Track*, Aug. 2021.
- [6] A. Curth and M. V. D. Schaar, "Nonparametric estimation of heterogeneous treatment effects: From theory to learning algorithms," in *Proc. Int. Conf. Artif. Intell. Statist.*, Jan. 2021, pp. 1810–1818.
- [7] R. H. Dehejia and S. Wahba, "Propensity score-matching methods for nonexperimental causal studies," *Rev. Econ. Statist.*, vol. 84, no. 1, pp. 151–161, Feb. 2002.
- [8] V. Dorie. (2016). *NPCI: Non-parametrics for Causal Inference*. [Online]. Available: <https://github.com/vdorie/npci>

- [9] M. Doutréline and G. Varoquaux, "How to select predictive models for decision making or causal inference," SSRN 4467871, France, Tech. Rep., 2023.
- [10] S. Feuerriegel, D. Frauen, V. Melnychuk, J. Schweisthal, K. Hess, A. Curth, S. Bauer, N. Kilbertus, I. S. Kohane, and M. van der Schaar, "Causal machine learning for predicting treatment outcomes," *Nature Med.*, vol. 30, no. 4, pp. 958–968, Apr. 2024.
- [11] J. H. Friedman and B. E. Popescu, "Predictive learning via rule ensembles," *Ann. Appl. Statist.*, vol. 2, no. 3, pp. 916–954, Sep. 2008.
- [12] F. Fumagalli, M. Muschalik, P. Kolpaczki, E. Hüllermeier, and B. Hammer, "SHAP-IQ: Unified approximation of any-order Shapley interactions," *Adv. Neural Inf. Process. Syst.*, Jan. 2023.
- [13] J. R. Harring, B. A. Weiss, and M. Li, "Assessing spurious interaction effects in structural equation modeling: A cautionary note," *Educ. Psychol. Meas.*, vol. 75, no. 5, pp. 721–738, Oct. 2015.
- [14] N. Hassanpour and R. Greiner, "CounterFactual regression with importance sampling weights," in *Proc. 28th Int. Joint Conf. Artif. Intell.*, Aug. 2019, pp. 5880–5887.
- [15] L. Henckel, E. Perković, and M. H. Maathuis, "Graphical criteria for efficient total effect estimation via adjustment in causal linear models," *J. Roy. Stat. Soc. Ser. B, Stat. Methodology*, vol. 84, no. 2, pp. 579–599, Apr. 2022.
- [16] J. L. Hill, "Bayesian nonparametric modeling for causal inference," *J. Comput. Graph. Statist.*, vol. 20, no. 1, pp. 217–240, Jan. 2011.
- [17] P. W. Holland, "Statistics and causal inference," *J. Amer. Stat. Assoc.*, vol. 81, no. 396, pp. 945–960, Dec. 1986.
- [18] F. Johansson, U. Shalit, and D. Sontag, "Learning representations for counterfactual inference," in *Proc. Int. Conf. Mach. Learn.*, Jan. 2016, pp. 3020–3029.
- [19] J. Kaddour, A. Lynch, Q. Liu, M. J. Kusner, and R. Silva, "Causal machine learning: A survey and open problems," 2022, *arXiv:2206.15475*.
- [20] D. Koller and N. Friedman, *Probabilistic Graphical Models: Principles and Techniques*. Cambridge, MA, USA: MIT Press, 2009.
- [21] S. R. Künzel, J. S. Sekhon, P. J. Bickel, and B. Yu, "Metalearners for estimating heterogeneous treatment effects using machine learning," *Proc. Nat. Acad. Sci. USA*, vol. 116, no. 10, pp. 4156–4165, Mar. 2019.
- [22] R. Lalonde, "Evaluating the econometric evaluations of training programs with experimental data," *Amer. Econ. Rev.*, vol. 76, no. 4, pp. 604–620, Jan. 1986.
- [23] X. Lin, G. Zhang, X. Lu, B. Han, K. Takeuchi, and H. Kashima, "Estimating treatment effects under heterogeneous interference," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discovery Databases*. Switzerland: Springer, Jan. 2023, pp. 576–592.
- [24] C. Louizos, U. Shalit, J. M. Mooij, D. Sontag, R. S. Zemel, and M. Welling, "Causal effect inference with deep latent-variable models," in *Proc. Adv. Neural Inf. Process. Syst.*, May 2017.
- [25] O. J. Maclaren and R. Nicholson, "What can be estimated? Identifiability, estimability, causal inference and ill-posed inverse problems," 2019, *arXiv:1904.02826*.
- [26] X. Nie and S. Wager, "Quasi-oracle estimation of heterogeneous treatment effects," *Biometrika*, vol. 108, no. 2, pp. 299–319, May 2021.
- [27] M. Oprescu, J. Dorn, M. Ghoumnia, A. Jesson, N. Kallus, and U. Shalit, "B-learner: Quasi-oracle bounds on heterogeneous causal effects under hidden confounding," 2023, *arXiv:2304.10577*.
- [28] Á. Parafita and J. Vitrià, "Estimand-agnostic causal query estimation with deep causal graphs," *IEEE Access*, vol. 10, pp. 71370–71386, 2022.
- [29] G. Parascandolo, N. Kilbertus, M. Rojas-Carulla, and B. Schölkopf, "Learning independent causal mechanisms," in *Proc. Int. Conf. Mach. Learn.*, Feb. 2018, pp. 4036–4044.
- [30] A. Paszke, S. Gross, S. Chintala, G. Chanan, E. Z. Yang, Z. DeVito, Z. Lin, A. Desmaison, L. Antiga, and A. Lerer, "Automatic differentiation in PyTorch," in *Proc. NIPS-W*, Oct. 2017.
- [31] J. Pearl, "[Bayesian analysis in expert systems]: Comment: Graphical models, causality and intervention," *Stat. Sci.*, vol. 8, no. 3, pp. 266–269, Aug. 1993.
- [32] J. Pearl, *Causality*. Cambridge, U.K.: Cambridge Univ. Press, 2009.
- [33] R. Pros and J. Vitrià, "Exploiting causal knowledge during CATE estimation using tree based metalearners," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discovery Databases*. Switzerland: Springer, Dec. 2024, pp. 261–276.
- [34] A. Rotnitzky and E. Smucler, "Efficient adjustment sets for population average causal treatment effect estimation in graphical models," *J. Mach. Learn. Res.*, vol. 21, no. 188, pp. 1–86, Jan. 2020.
- [35] B. Schölkopf, F. Locatello, S. Bauer, N. R. Ke, N. Kalchbrenner, A. Goyal, and Y. Bengio, "Toward causal representation learning," *Proc. IEEE*, vol. 109, no. 5, pp. 612–634, May 2021.
- [36] U. Shalit, F. Johansson, and D. Sontag, "Estimating individual treatment effect: Generalization bounds and algorithms," in *Proc. 34th Int. Conf. Mach. Learn.*, D. Precup and Y. W. Teh, Eds., Jan. 2016, pp. 3076–3085.
- [37] C. Shi, D. M. Blei, and V. Veitch, "Adapting neural networks for the estimation of treatment effects," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, Jan. 2019.
- [38] S. Shimizu, P. O. Hoyer, A. Hyvärinen, A. Kerminen, and M. Jordan, "A linear non-Gaussian acyclic model for causal discovery," *J. Mach. Learn. Res.*, vol. 7, no. 10, pp. 1–28, 2006.
- [39] G. Tesei, S. Giampanis, J. Shi, and B. Norgéot, "Learning end-to-end patient representations through self-supervised covariance balancing for causal treatment effect estimation," *J. Biomed. Informat.*, vol. 140, Apr. 2023, Art. no. 104339.
- [40] M. Thoresen, "Spurious interaction as a result of categorization," *BMC Med. Res. Methodology*, vol. 19, no. 1, pp. 1–8, Dec. 2019.
- [41] A. Whata and C. Chimedza, "Evaluating uses of deep learning methods for causal inference," *IEEE Access*, vol. 10, pp. 2813–2827, 2022.
- [42] K. Xia, K.-Z. Lee, Y. Bengio, and E. Bareinboim, "The causal-neural connection: Expressiveness, learnability, and inference," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, Jan. 2021, pp. 10823–10836.
- [43] J. Yoon, J. V. Jordan, and M. V. D. Schaar, "GANITE: Estimation of individualized treatment effects using generative adversarial nets," in *Proc. Int. Conf. Learn. Represent.*, Feb. 2018.
- [44] K. Yu, X. Guo, L. Liu, J. Li, H. Wang, Z. Ling, and X. Wu, "Causality-based feature selection: Methods and evaluations," *ACM Comput. Surveys*, vol. 53, no. 5, pp. 1–36, Sep. 2021.
- [45] K. Zhang, S. Zhu, M. Kalandar, I. Ng, J. Ye, Z. Chen, and L. Pan, "GCastle: A Python toolbox for causal discovery," 2021, *arXiv:2111.15155*.



ROGER PROS received the B.S. degree in statistics from the Universitat de Barcelona (UB), Spain, in 2016, and the M.S. degree in industrial and applied mathematics from Grenoble INP Ensimag, France, in 2017. He is currently pursuing the Ph.D. degree in mathematics and informatics with UB.

With a strong background in data science and analytics, he was a Data Scientist with leading organizations, such as Boston Consulting Group and PepsiCo, where he applied advanced statistical and machine learning techniques to solve complex business challenges. His research interests include causal inference and causal machine learning, aiming to develop robust methodologies for understanding cause-and-effect relationships in data-driven environments.



JORDI VITRIÀ joined the Universitat de Barcelona (UB), in 2007, where he is currently a Full Professor, where he teaches an introductory course on algorithms for data analysis and advanced courses on data science, causality, and deep learning. From April 2011 to January 2016, he was the Head of the Applied Mathematics and Analysis Department, UB, where he is also a member of the Mathematics and Computer Science Department. His research, when personal computers had 128 KB

of memory, was originally oriented toward digital image analysis and how to extract quantitative information from them, but soon evolved towards computer vision problems. After a postdoctoral year with the University of California at Berkeley, in 1993, he focused on Bayesian methods for computer vision methods. He is leading a research group working in causal inference and machine learning. He has authored more than 100 peer-reviewed articles and holds eight international patents. He has directed 15 Ph.D. theses in the area of machine learning and computer vision. He has been the leader of a large number of research projects at international and national level.

...